Far Side Moon Rocks Help Explain the Moon's Two Faces

From: ScienceDaily Top Technology Mined: 2026-10-15 5 min read

The side of the Moon that faces Earth looks very different from the side that always faces away. Now scientists in China think they have found new clues that help explain why.

The near side has huge, dark plains made of cooled volcanic rock called **basalt**. The far side is mostly old, rough highlands covered in craters, with a much thicker **crust**. Scientists have argued for decades about what caused this difference.

In 2024, China's Chang'e-6 mission became the first to bring back rock samples from the Moon's far side. A team led by Professor Hejiu Hui of Nanjing University studied the chemistry of those rocks. They compared them with rocks from the near side collected by the Apollo missions and China's Chang'e-5 mission.

The team found something surprising. Deep inside the Moon, in a layer called the **mantle**, the material on both sides seems to be very similar. But the **crust**, the Moon's outer layer, is clearly different. Far side rocks held more of the element magnesium and very small amounts of **rare earth elements**.

The researchers suggest an explanation. Long ago, the young Moon was covered by an ocean of melted rock, called a **magma ocean**. At first, it cooled evenly. Later, the Moon became "**tidally locked**," meaning the same side always faced Earth. At that time, Earth and the Moon were much closer together. Earth's gravity pulled on the Moon and created more heat on the near side than on the far side.

According to the team's model, this uneven heating stirred huge flows of hot, melted rock inside the Moon. Over time, those flows helped give the Moon its two very different faces.

Word Treasure

basalt -- Basalt is a dark rock that forms when hot lava cools and hardens.

crust -- The crust is the hard, rocky outer layer of a planet or moon.

mantle -- The mantle is the thick layer of rock that lies under the crust.

rare earth elements -- Rare earth elements are a group of metals that scientists use to learn how rocks formed.

magma ocean -- A magma ocean is a huge layer of melted rock that covered a young planet or moon.

tidally locked -- Tidally locked means one side of a moon always faces the planet it circles.

Background you need

The Moon spins exactly once each time it circles Earth, so the same side always faces us and the far side can only be seen by spacecraft.

Scientists learn how the Moon formed by studying the chemistry of its rocks, which can show where the material came from and how it cooled.

Most Moon rocks studied in labs came from the near side, collected by the Apollo missions and China's Chang'e-5 mission.

China's Chang'e-6 mission was the first to bring back samples from the far side, letting scientists compare both halves directly.

Break it down

LEAD: Scientists in China find new clues about why the Moon's two sides look so different.
PUZZLE: Near side has dark volcanic plains; far side has cratered highlands and thicker crust
KEY EVENT: Study of far-side rocks from Chang'e-6
+- WHO: Team led by Professor Hejiu Hui, Nanjing University
+- HOW: Compared with near-side rocks from Apollo and Chang'e-5
+- WHEN: Samples returned in 2024
EVIDENCE: Mantle similar on both sides; crust different, with more magnesium and few rare earth elements on the far side
EXPLANATION: After tidal locking, Earth's gravity heated the near side more, stirring flows of melted rock
OPEN QUESTION: The model is the team's proposed explanation for a long-debated mystery.

Why it matters

Understanding why the Moon has two faces helps scientists learn how the Moon formed and changed, and how Earth's gravity shaped its closest neighbor.

Test what you remember

1. What covers much of the Moon's near side?

- (A) Deep oceans of water
- (B) Green forests
- (C) Thick ice
- (D) Huge, dark plains of cooled volcanic rock

2. Which mission was the first to bring back samples from the Moon's far side?

- (A) Apollo 11
- (B) Voyager 1
- (C) Chang'e-5
- (D) Chang'e-6

3. Who led the research team?

- (A) An Apollo astronaut
- (B) Professor Hejiu Hui of Nanjing University
- (C) Professor Bohm of Oxford
- (D) A NASA rover

4. What did the team find about the mantle on both sides?

- (A) It seems very similar
- (B) It is completely missing
- (C) It is much hotter on the far side today
- (D) It is made of ice

5. What does "tidally locked" mean in the article?

- (A) Earth's oceans froze
- (B) The same side of the Moon always faces Earth
- (C) The Moon has no tides
- (D) The Moon stopped moving

6. According to the team's model, what made the near side hotter long ago?

- (A) Earth's gravity pulling on the Moon when they were closer
- (B) Meteors hitting the far side
- (C) Volcanoes on Earth
- (D) Sunlight only

Answer key (try the quiz first!): 1: D 2: D 3: B 4: A 5: B 6: A

Questions to think about

MY THOUGHTS (at least 20 words):